Exp -4 SETTING UP A LOCAL AREA NETWORK (LAN)

8/8/26

Aim: To design, set up and configure a local area network (LAN) connecting 4 computers through a network switch and demonstrate resource/data sharing across the network.

Equipment Required:

Equipment	Quantity	specification
Powered Computer	4	Ethernet (NIC) Enabled
Network switch	1	8/16/24 port
Ethernet cables	4	straight through RJ-45 connectors

IP address configuration table:

Device	IP Address	Subnet mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch port Fa 0/1
PC1	10.1.1.2	255.0.0.0	Switch port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch port Fa 0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation in CISCO Packet Tracer

1) Place the devices:

- open application, choose End Devices and drag 4 PCs into workspace
- Select switches and drag it in

2) Cable the device:

- Select copper straight through cable in connections.
- Click PC, then switch then fast ethernet port for each PC.
- Confirm successfull connection via green link lights.

3) Assign IP addresses to PC:

- click PC -> Desktop Tab - IP configuration
- Enter static IPv4 address and subnet mask as per IP table.

4) Configure switch management IP.

- ~~Click~~ switch -> CLI tab.
  -> enable.

- Interface VLAN
  ip address 10.1.1.5 255.0.0.0
  no shutdown
  exit.

5) Verify connectivity using ping

- click a PC → Desktop tab → prompt
- ping (ip of ~~source~~ ~~ip of~~ destination)

6) Demonstrate Data Sharing (server simulation)

- Add server from End devices → server.
- Connect it with switch, assign IP
- enable HTTP service for server.
- use another PC's web browser and enter server's IP address to load the page.

Result:-

Thus, LAN is setup and connection is verified successfully and also implemented in Cisco Packet Tracer.